CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Fourth Semester of B. Tech. Theory Examination (IT)
University Examination - May 2014
MA-206: Probability, Statistics and Numerical Analysis

Date: 09.05.2014, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator (non-programmable) is allowed.

SECTION – I

Q - 1 Write whether following statement are true or false [07]

1. Normal probability distribution is a type of discrete probability.
2. In a distribution of 20, 30, 40, the $\bar{x}$ is 30, the sum of deviations from x will be zero.
3. The standard deviation of 5, 5, 5, 5 and 5 is zero.
4. The value of Karl-Pearson Coefficient of correlation r satisfies $0 \le r \le 1$
5. If the mean of Binomial distribution is 1.50 and p = 0.2 then number of trial n = 4.
6. If A & B are independent events then probability function satisfies $P(A \cap B) = P(A)P(B)$
7. Binomial distribution is a limiting form of Poisson distribution.

Q - 2 (a) The following measurements of the specific heat of a certain chemical were made in order to investigate the variation in specific heat with temperature: [07]

Temp ^{0}C	0	10	20	30	40	50
Specific Heat	0.51	0.55	0.57	0.59	0.63	0.67

Estimate the regression line of specific heat on temperature, and predict the value of the specific heat when the temperature is 65 ^{0}C.

(b) The following data give the experience of machine operators and their performance ratings as given by the number of good parts tuned out per 100 pieces: Find Karl Pearson coefficient of correlation between Experience and Performance ratings and interpret it. [07]

Experience (in years) (x)	16	12	18	4	3	10	5	12
Performance ratings (y)	87	88	89	68	78	80	75	83

OR

Q - 2 (a) Write properties of correlation and regression coefficient. [04]

(b) find the arithmetic mean for the distribution given below: [05]

Class	10-20	20-30	30-40	40-50	50-60	60-70	70-80	Total
Frequency	42	38	98	54	100	36	32	400

(c) The following frequency distribution shows the numbers of defects in each lot in a production unit. Find the median and mode. [05]

No of Defects	0	1	2	3	4	5	6
No of lots	6	30	42	55	25	18	5

Q - 3 (a) Write properties of Arithmetic mean. [05]

(b) Subway trains on a certain line run every half hour between mid-night and six in the morning. What is the probability that a man entering the station at a random time during this period will have to wait at least twenty minutes? [05]

(c) The problem in mathematics is given to three students A, B and C whose chances of solving it are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ respectively. What is the probability that the problem will be solved? [04]

OR

Q - 3 (a) The linear measurement of the items of a products are approximately normally distributed with a mean of 20 cm and standard deviation 4 cm. Items which measure between 18 cm and 23 cm are sold at Rs. 50 each and the other items at Rs. 30 each. Find the total amount collected if in all 10,000 items are sold. How many items are of measurement 26 cm or more? [05]

(b) The probability of non-happening an event A is 0.3 and the probability of happening an event B is 0.5. The probability of happening at least one of the two events A and B is 0.8. Find the probability of happening both the events A and B. what is the probability that none of the events will happen? [05]

(c) If 5% of the electric bulbs manufactured by a company are defective, use Poisson distribution to find the probability that in a sample of 100 bulbs (i) none is defective (ii) 5 bulls will be defective. [04]

SECTION – II

Q - 4 Answer following questions [07]

1. Simpson 1/3 Rule is applicable when number of sub interval is 7. (True/False)
2. Truncation error is the error occurs due to truncating a number. (True/False)
3. Write general equation of hyperbola.
4. Lagrange's method is applicable when data is equally spaced. (True/False)
5. Rate of convergence for Bisection method is 0.5. (True/False)
6. Runge-Kutta method of order 2 gives solution as function relationship between variables. (True/False)
7. Write Picard's formula, to find solution of linear ordinary differential equation of one order.

Q - 5 (a) Use Lagrange's Interpolation formula to find interpolated value of Y for X = 2 [05]

X	-1	0	1	2
Y	3	1	3	7

(b) Use appropriate Interpolation formula to find interpolated value of Y for X = 7 [05]

X	2	4	6	8	10
Y	9	65	217	512	1000

(c) Define relative error. Find relative error if Π is approximated correct up to four decimal places. [04]

OR

Q - 5 (a) Use Trepazoidal Rule to evaluate integral $y = \int_0^1 \frac{1}{1+x^2}\,dx$, using 7 sub-intervals. [05]

(b) Use Simpson's 1/3 Rule to evaluate integral $y = \int_0^2 \sin(2x+7)\,dx$, use 6 sub-intervals. [05]

(c) Growth rate of bacteria (g in thousands) with respect to time (t in secs.) given below, find best fitted geometric curve to the given data: [04]

t	10	20	30	40	50
g	150	200	280	390	500

Q - 6 (a) Apply Runge-Kutta method of order 2 find $y(0.1), y(0.2)$ provided that, [05]

$$\frac{dy}{dx} = y - x \text{ with } y(0) = 2$$

(b) Apply Picard's Rule to find second approximation of solution, for differential equation $y' = x + y^2;\ y(0) = 1$ [05]

(c) Given that a number x is approximated as 3.45, and relative error is not more than 2% then find an interval within which exact value of x lies. [04]

OR

Q - 6 (a) Find positive root of $x^2 - 2x - 3 = 0$ by Bisection method, using initials 2.5 and 5. [05]

(b) Solve equation $x^3 - 125 = 0$ using Newton Raphson method, using appropriate initial. [05]

(c) Solve $e^x = \cos x$ using successive approximation method. Use appropriate initial. [04]

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